

Project: Zero-to-Brand

Architecture

- **Frontend:** Next.js 16 App Router (Tailwind CSS v4).
- **Backend:** API Routes (Next.js serverless functions) calling Gemini 2.5 Flash and Imagen 3.0 via `@google/genai`.
- **Database:** Local JSON file-based database (`src/data/db.json`) managed by a custom client (`src/data/dbClient.ts`).

Milestones

#	NAME	SCOPE	DEPENDENCIES	STATUS
1	E2E Test Suite Setup	Build E2E test infra and test cases (Tiers 1-4) in a test runner	None	PLANNED
2	Brand Onboarding Survey Expansion (R1)	Expand survey (brand owner, target audience, preferred design styles) and persist preferences in JSON database	None	PLANNED
3	Image Questionnaire & Modular Agents Backend (R2)	Add follow-up questionnaire on upload and implement the three modular under-the-hood agents (Preferences, Prompt Engineering, Designer)	M2	PLANNED
4	Dashboard UI & Canvas Poster Compositor (R3)	Build workspace showing craft, studio, and composite poster. Implement frontend canvas poster composition, saving, and downloading	M3	PLANNED
5	Integrated E2E Verification & Review	Pass 100% of the E2E test suite and run adversarial coverage hardening (Tier 5)	M1, M4	PLANNED

Interface Contracts

Brands API (`/api/brands`)

- **GET:** Retrieves a specific brand or list of all brands.

- **POST:** Creates a draft brand from a craft image.
- **PUT:** Updates brand variables, chat history, or onboarding config (owner, audience, dials, design styles).
- **DELETE:** Removes a brand.

Products API (`/api/products`)

- **GET:** Retrieves all products associated with a brand.
- **POST:** Takes brandId, rawImage, and image-specific questionnaire answers (desired scene, marketing style, keywords). Returns created product.
- **PUT:** Updates product details, aspect ratio, style preset, or regenerates banner using feedback.
- **DELETE:** Removes a product.

Chat API (`/api/gemini/chat`)

- **POST:** Handles chat conversation between user and Brand Co-pilot.

Ingest API (`/api/gemini/ingest`)

- **POST:** Performs initial Gemini analysis of a brand's craft photo.

Finalize API (`/api/gemini/finalize`)

- **POST:** Runs Imagen 3 to generate brand logo and initial banner, completes brand onboarding.

Code Layout

- `src/app/` - Next.js page components and API routes
- `src/lib/` - Shared services, Gemini utility, and modular agent implementations
- `src/data/` - Database client and JSON database
- `tests/` - E2E and unit test suite